

Practical 7: Time Series Decomposition (Additive vs. Multiplicative)

Question: Using Python, decompose a time series into its Trend, Seasonality, and Residual components using additive decomposition. Explain whether additive or multiplicative decomposition is more suitable for the chosen dataset and justify your answer.

7.1 Objective

- To decompose a time series using the Additive model.
- To compare it with the Multiplicative model.
- To determine which model fits the data better based on the residual patterns.

7.2 Theory

- **Additive Model ($Y_t = T_t + S_t + R_t$):** Assumes the seasonal amplitude remains constant regardless of the trend. Useful when seasonal variations are roughly the same size over time.
- **Multiplicative Model ($Y_t = T_t \times S_t \times R_t$):** Assumes the seasonal amplitude changes (grows or shrinks) relative to the trend. Useful when peaks/troughs widen as the level of the series increases.

7.3 Code Implementation

```
# 1. Load Data
# Using AirPassengers (classic example of multiplicative seasonality)
dataset = sm.datasets.get_rdataset("AirPassengers")
ts_data = dataset.data['value']
ts_data.index = pd.date_range(start='1949-01-01', periods=len(ts_data), freq='MS')

# 2. Perform Additive Decomposition
decomp_add = seasonal_decompose(ts_data, model='additive')

# 3. Perform Multiplicative Decomposition
decomp_mult = seasonal_decompose(ts_data, model='multiplicative')

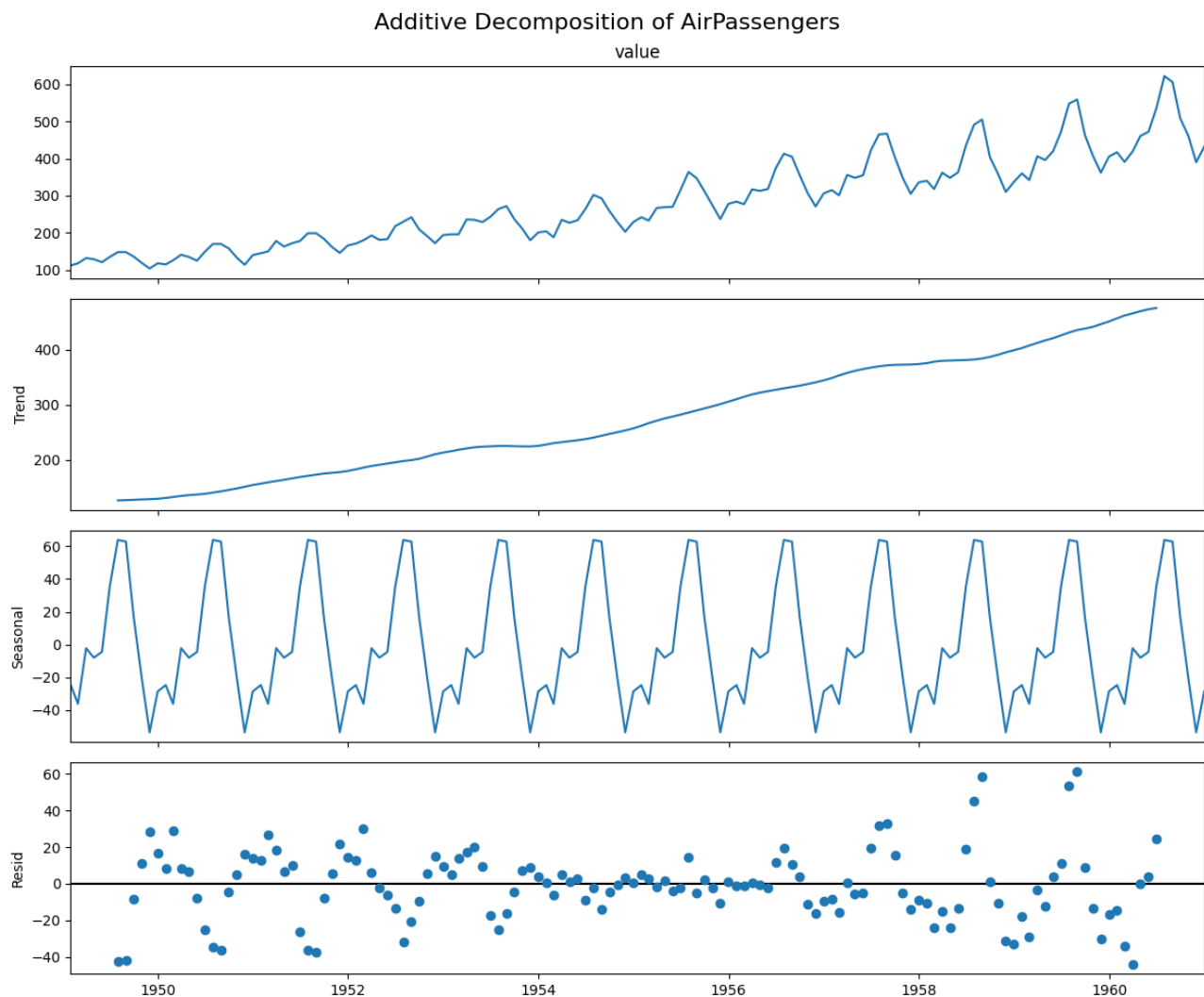
# 4. Plot Comparison
fig, axes = plt.subplots(2, 1, figsize=(12, 10))

# Plot Additive Residuals
decomp_add.resid.plot(ax=axes[0], title='Additive Model Residuals')
axes[0].set_ylabel('Residuals')

# Plot Multiplicative Residuals
decomp_mult.resid.plot(ax=axes[1], title='Multiplicative Model Residuals', color='green')
axes[1].set_ylabel('Residuals')

plt.tight_layout()
```

`plt.show()`



7.4 Interpretation & Justification

- **OBSERVATION:**
 - The **Additive Decomposition** residuals show a distinct pattern (larger errors at the beginning and end, or "bowing"), indicating that the model failed to capture the changing magnitude of seasonality.
 - The **Multiplicative Decomposition** residuals appear more random and uniform (close to white noise) with no obvious pattern.
- **Conclusion: Multiplicative decomposition** is more suitable for this dataset.
- **Justification:** The dataset exhibits "expanding" seasonality—the difference between the peaks and troughs increases as the trend moves upward. The additive model forces a constant seasonal swing, leaving the excess variation in the residuals, whereas the multiplicative model scales the seasonality with the trend, fitting the data correctly.